Problem 61. A sequence $a_0, a_1, a_2, \ldots$ of positive integers satisfies the following recurrence relation:

$$a_{n+1} = \begin{cases} \frac{1}{3}a_n, & \text{if } 3 \mid a_n, \\ 2a_n + 1, & \text{otherwise,} \end{cases}$$

for all integers $n \geqslant 0$.

Show that, as n increases to infinity, one of the following cases must hold:

Case 1: Either the sequence $\{a_n\}$ increases to infinity, or

Case 2: $\{a_n\}$ ends up in an alternating cycle $1, 3, 1, 3, 1, \ldots$.

Solution 1. Consider the following possibilities:

(1) if $a_k = 1$ for some non-negative integer k, then $a_{k+1} = 2 \cdot 1 + 1 = 3$, $a_{k+2} = \frac{1}{3} \cdot 3 = 1, \ldots$. So the sequence eventually ends up in an alternating cycle $1, 3, 1, 3, \ldots$ and Case 2 holds;

(2) if $a_n \equiv -1 \pmod 3$ for some non-negative integer n, then, by induction, $a_k \equiv -1 \pmod 3$ for all $k \geqslant n$, and so $a_{k+1} - a_k = a_k + 1 > 0$. Hence we have an increasing sequence and Case 1 holds.

Now we claim that all sequences fall in either of the two above possibilities. The proof proceeds by contradiction. Suppose that we have a sequence where

- $a_n \geqslant 2$ (because it is not (1)) and
- $a_n \not\equiv -1 \pmod 3$ (because it is not (2)),

for all non-negative integers n.

Now observe that

- if $a_n \equiv 0 \pmod 3$, then $a_{n+1} = \frac{1}{3}a_n < a_n$;
- if $a_n \equiv 1 \pmod 3$, then $a_{n+1} = 2a_n + 1 \equiv 0 \pmod 3$ so $a_{n+2} = \frac{1}{3}(2a_n + 1) < a_n$ (because $a_n > 1$).

So to produce a contradiction we need to construct a strictly decreasing infinite subsequence of the sequence $\{a_n\}$ of positive integers. One way to do this is to take the original sequence and delete all terms that are multiples of 3. Thus we have a strictly decreasing infinite subsequence of positive integers, which is impossible!

Therefore all sequences fall in either of the two above possibilities and so either Case 1 or Case 2 holds. □

Solution 2. A more elegant way is to solve this problem without constructing an infinite subsequence.

Consider the index k such that the term a_k is the smallest value (over all a_n, where n is a non-negative integer). If the same smallest value is attained multiple times, we pick k arbitrarily among the minimal values. Now we look at the three possible residue classes for $a_k \pmod 3$.

- If $a_k \equiv -1 \pmod 3$, then (as shown above) the sequence increases to infinity from then on and Case 1 holds.
- If $a_k \equiv 0 \pmod 3$, then $a_{k+1} = \frac{1}{3}a_k < a_k$ and this contradicts the minimality of a_k and so in this case there is no such k.
- If $a_k \equiv -1 \pmod 3$, then $a_{k+1} = 2a_k + 1 \equiv 0 \pmod 3$. Hence $a_{k+2} = \frac{1}{3}(2a_k + 1) \geqslant a_k$ (because a_k is minimal) and so $a_k \leqslant 1$. As a_k is a positive integer, this forces $a_k = 1$ and so the sequence ends up in an alternating cycle $1, 3, 1, 3, \ldots$ and Case 2 holds. □

Problem 62. Find all triples (a, b, c) of positive integers such that

$$\begin{cases} a^2 + b^2 = n \operatorname{lcm}(a, b) + n^2, \\ b^2 + c^2 = n \operatorname{lcm}(b, c) + n^2, \\ c^2 + a^2 = n \operatorname{lcm}(c, a) + n^2, \end{cases}$$

for some positive integer n.

Solution. The answers are all triples $(a, b, c) = (k, k, k)$, where k is any positive integer.

It can be easily checked that all such triples are solutions where $n = k$. Conversely, suppose that (a, b, c) is a solution. We then need to show that $a = b = c$.

Suppose that there exists some integer $d > 1$ such that $d \mid a$, $d \mid b$ and $d \mid c$. From any of the equations of the system (by using the quadratic equation to solve for n), we also get $d \mid n$. Thus, by replacing (a, b, c) with $(a/d, b/d, c/d)$, we obtain a new solution, where n is replaced by

n/d. Thus, without loss of generality, we can assume that a, b, c, and n share no common divisor other than 1. Now subtract the second equation from the first equation and add this to third equation to obtain

$$2a^2 = n(\operatorname{lcm}(a,b) - \operatorname{lcm}(b,c) + \operatorname{lcm}(c,a) + n).$$

Hence $n \mid 2a^2$, and similarly, $n \mid 2b^2$, and $n \mid 2c^2$. But as a, b, c and n share no common divisor other than 1, it then follows that either $n = 1$ or $n = 2$. If $n = 1$, then we have

$$2ab \leqslant a^2 + b^2 = \operatorname{lcm}(a,b) + 1 \leqslant ab + 1.$$

This gives $a = b = c = 1$, which leads to the family of solutions (k, k, k). If $n = 2$, then

$$a^2 + b^2 = 2\operatorname{lcm}(a,b) + 4 \leqslant 2ab + 4$$

and so $|a - b| \leqslant 2$. Similarly, $|b - c| \leqslant 2$ and $|c - a| \leqslant 2$. Note that no two of a, b, and c can be consecutive. To see this, suppose, without loss of generality, that $a = b + 1$. Substituting this to the first equation, we have $(b + 1)^2 + b^2 = 6$, which is impossible. Thus, at least two of a, b, and c must be equal. Without loss of generality, assume that $a = b$. Substituting to the first equation, we obtain $a = b = 2$. Thus, $4 + c^2 = 2\operatorname{lcm}(2, c) + 4$, and so c is even. This is a contradiction since we assumed that a, b, c, and n have no common divisor other than 1. Therefore, the case $n = 2$ does not give any additional solution. □

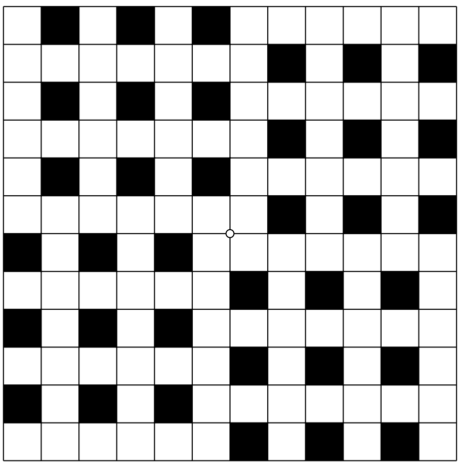
Problem 63. Let $k > 1$ be a positive integer. Find the smallest positive integer n such that some cells of an $n \times n$ grid of unit squares can be coloured black in such a way that each row and each column contains exactly k black cells and no two black cells have a common side or a common vertex.

Solution. The answer is $n = 4k$ where $k > 1$ is the given positive integer.

In every 2×2 square a maximum of one cell can be black. Let us consider any two consecutive rows in the grid. They must contain exactly $2k$ black cells in total. By dividing the cells of the two rows into 2×2 squares starting from the left, we see that the number of 2×2 squares must be at least $2k - 1$ (the last black cell can be alone in the rightmost column). Thus $n \geqslant 2 \cdot (2k - 1) + 1 = 4k - 1$.

Let us assume that $n = 4k - 1$. Out of the black cells in any two consecutive rows, at most $2k - 1$ can be in the leftmost $4k - 2$ columns. Thus one black square must be located in the last column. But then the second to last column cannot have any black squares in it. As this holds for any two consecutive rows of an $n \times n$ grid, the second to last column of the entire grid cannot contain any black cells. This, however, contradicts the assumption that each column contains k black cells. Thus $n \geqslant 4k$.

One suitable construction for $n = 4k$ is the following: take four grids of size $2k \times 2k$, in which exactly those cells that are in odd rows and even columns have been coloured black, and place them around a point by rotating them $90°$ with respect to each other. The case $n = 12$ with $k = 3$ is shown below.



□

Problem 64. Samuel and Mandy play the following game. Samuel starts by drawing a random triangle on a piece of paper. Mandy then draws a line on the same paper that goes through the midpoint of one of the sides of the medial triangle of the triangle. Then Samuel adds another line that also goes through the midpoint of the same side of the medial triangle of the triangle. These two lines divide the triangle into four pieces. Samuel gets the piece with maximum area (or one of those with maximum area) and the piece with minimum area (or one of those with minimum area), while Mandy gets the other two pieces. The player whose total area is bigger wins. Does either of the players have a winning strategy, and if so, who has it?

Solution. The answer is no.

The midpoint of a side of the medial triangle of the triangle is located on the median of the triangle. Indeed, let D, E and F be the midpoints of sides BC, CA and AB of $\triangle ABC$, respectively and let K be the point of intersection of median AD and EF. Since $\triangle AEK \sim \triangle ACD$ (because $EK \parallel CD$), we have

$$\frac{EK}{EF} = \frac{EK}{CD} = \frac{AE}{AC} = \frac{1}{2}.$$

that is, the median bisects EF of the medial triangle of the triangle.

Mandy can avoid loss by drawing the extension of a median of the triangle during her move. This move respects the rules. The line divides the triangle into two regions with the same area. Let Samuel's second move divide the first of these regions into two pieces with areas S_1 and S_2 and the second one into two pieces with areas S_3 and S_4. Without loss of generality, let S_1 be the greatest area among these four. Since $S_1 + S_2 = S_3 + S_4$, the area S_2 must be the least among these areas. Thus Samuel gets precisely half of the original triangle and Mandy gets the second half. Samuel can also avoid loss. If Mandy draws the extension of the median during her move, then based on the previous part neither player wins. If Mandy draws any other line through the midpoint of some side of the medial triangle, then Samuel can draw the median himself and thus both again get an equal total area. Therefore neither of the two has a winning strategy. $\square$